

Construction of the Lebesgue Measure

A collection of subsets of X , $A \subset 2^X$, is called an **algebra** on X if: (1) $\emptyset \in A$, (2) A is closed under complements, (3) A is closed under finite unions.

$A \subset 2^X$ is called a **σ -algebra** on X if it is an algebra on X and closed under countable unions. The smallest σ -algebra on $\mathbb{R}$ containing all open subsets of $\mathbb{R}$ is called the **Borel σ -algebra**, denoted as $\mathcal{B}(\mathbb{R})$. An element of this σ -algebra is called a **Borel set**.

Definition: Measure Function

Suppose X is a set and Σ is a σ -algebra on X . A **measure** on (X, Σ) is a function $\mu : \Sigma \rightarrow [0, \infty]$ such that $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k)$$

for every disjoint sequence $E_1, E_2, \dots$ of sets in Σ .

Definition: Lebesgue Measure

- If E is any subset of $\mathbb{R}^n$, the **outer measure** of E is the function $\mathcal{L}^n : 2^{\mathbb{R}^n} \rightarrow [0, \infty]$ defined as

$$\mathcal{L}^n(E) = \inf \sum_{j=1}^{\infty} |Q_j|$$

where the infimum is taken over all countable coverings $E \subset (\bigcup_{j=1}^{\infty} Q_j)$ of closed cubes.

- The **Lebesgue measure** is the restriction of $\mathcal{L}^n$ to $\mathcal{B}(\mathbb{R})$, becoming a measure on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ that assigns each Borel set of $\mathbb{R}$ to its outer measure.

Hausdorff Measure & Hausdorff Dimension

Definition: Hausdorff Measure

Let $A \subseteq \mathbb{R}^n$ and $s \geq 0$. For $\delta > 0$, let $\mathcal{H}_{\delta}^s(A)$ be the following:

$$\mathcal{H}_{\delta}^s(A) = \inf \left\{ \sum_{i=1}^{\infty} (\text{diam}(E_i))^s : A \subseteq \bigcup_{i=1}^{\infty} E_i, \text{diam}(E_i) \leq \delta \text{ for all } i \in \mathbb{N} \right\}.$$

Then the **s -dimensional Hausdorff outer measure** of A is given by

$$\mathcal{H}^s(A) = \lim_{\delta \rightarrow 0} \mathcal{H}_{\delta}^s(A) = \sup_{\delta > 0} \mathcal{H}_{\delta}^s(A).$$

Relation to the Lebesgue Measure: If $A \subseteq \mathbb{R}^n$, then $\mathcal{H}^n(A) = c_n \mathcal{L}^n(A)$, where $c_n = \pi^{\frac{n}{2}} / 2^n \Gamma(\frac{n}{2} + 1) = \mathcal{L}^n(B_{\frac{1}{2}}(0))$, the volume of the n -dimensional ball.

Scaling Property: Let $F \subseteq \mathbb{R}^n$ and $\lambda > 0$, then $\mathcal{H}^s(\lambda F) = \lambda^s \mathcal{H}^s(F)$

Regularity: Let $A \subseteq \mathbb{R}^n$ be a subset. Then: (1) there is a G_{δ} -set G such that $A \subseteq G$ and such that $\mathcal{H}^s(A) = \mathcal{H}^s(G)$, (2) if A is $\mathcal{H}^s$ -measurable and $\mathcal{H}^s(A) < \infty$, then there is an F_{σ} -set F such that $F \subseteq A$ and such that $\mathcal{H}^s(A) = \mathcal{H}^s(F)$.

Definition: Hausdorff Dimension

Let $F \subseteq \mathbb{R}^n$. Then there exists a unique value α such that for all $s < \alpha$, we have $\mathcal{H}^s(F) = \infty$, and for all $s > \alpha$, we have $\mathcal{H}^s(F) = 0$. We call α the **Hausdorff dimension** of F , which we denote by $\dim(F)$.

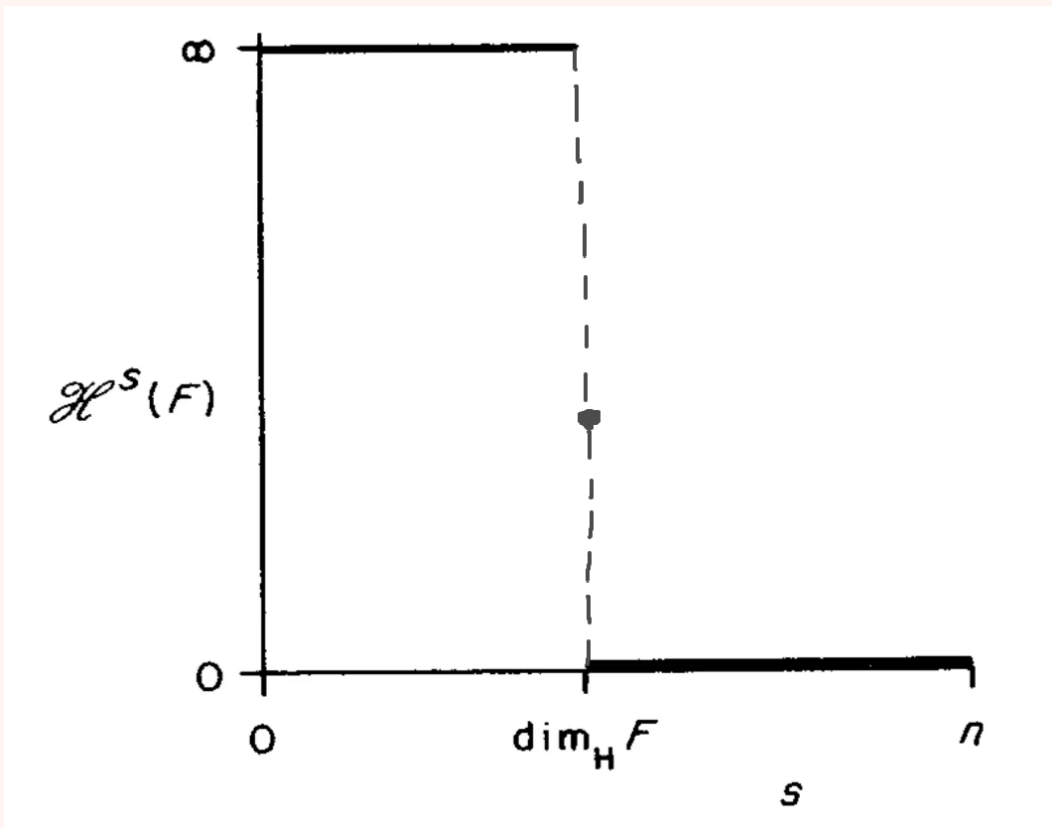


Figure 1. Graph of $\mathcal{H}^s(F)$ with respect to s

Basic Terminologies

A **akeya set** is a set in $\mathbb{R}^n$ that contains a line segment of unit length in every direction.

A **akeya needle set** is a set in $\mathbb{R}^n$ such that a unit line segment can be translated and rotated back to its original position within the set.

Minimal Area ofakeya Sets

A triangle T of height h can be transformed into T' in the following manner: by cutting through its apex, and translating by $1 - \alpha$, the new figure formed, F , would be such that the area is reduced by $(1 - \alpha)(3\alpha - 1)\mathcal{L}^2(T)$, as by similar triangles, $\mathcal{L}^2(t) = \alpha^2 \mathcal{L}^2(T)$, and $\mathcal{L}^2(A) = \mathcal{L}^2(D) = \mathcal{L}^2(B) = \mathcal{L}^2(C)$ and each is equal to $\frac{1}{2}(1 - \alpha)^2 \mathcal{L}^2(T)$.

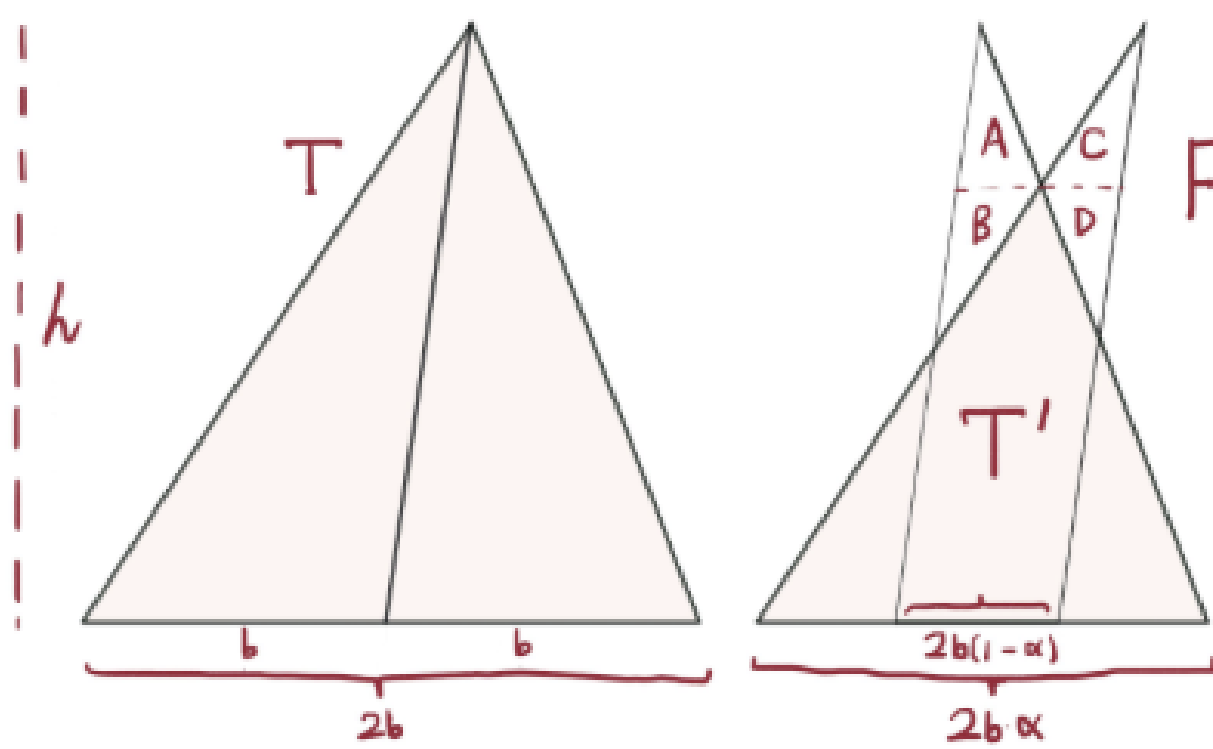


Figure 2. Transformation from triangle T to figure F

Lemma: Transformation of Triangles

Let $\varepsilon > 0$ and T be a triangle. Applying the above procedure repeatedly, we can make the resulting figure F satisfy $\mathcal{L}^2(F) < \varepsilon$.

A triangle T can be turned into a Perron Tree by iterating this: by subdividing the base of the triangle into 2^k segments and applying the transformation repeatedly on adjacent triangles, a new figure can be created with a reduced measure, while still preserving a line of every direction between $\pi/3$ and $2\pi/3$. The full process takes k steps, and at the j^{th} step, the reduction is bounded below by $(1 - \alpha)(3\alpha - 1)\alpha^{2(j-1)} \mathcal{L}^2(T)$. Hence, by summing over all the steps,

$$\sum_{i=1}^{2^{k-1}} \mathcal{L}^2(T_{2i-1} \cup T_{2i}) - \mathcal{L}^2(F_1^{(k)}) \geq (1 - \alpha)(3\alpha - 1)(1 + \alpha + \alpha^2 + \dots + \alpha^{2(k-1)}) \mathcal{L}^2(T)$$

and

$$\mathcal{L}^2(F) \leq \left(1 - \frac{(3\alpha - 1)(1 - \alpha^{2k-1})}{1 + \alpha}\right) \mathcal{L}^2(T).$$

Therefore, we can make the resultant figure F satisfy $\mathcal{L}^2(F) < \varepsilon$ by an appropriate choice of α and k .

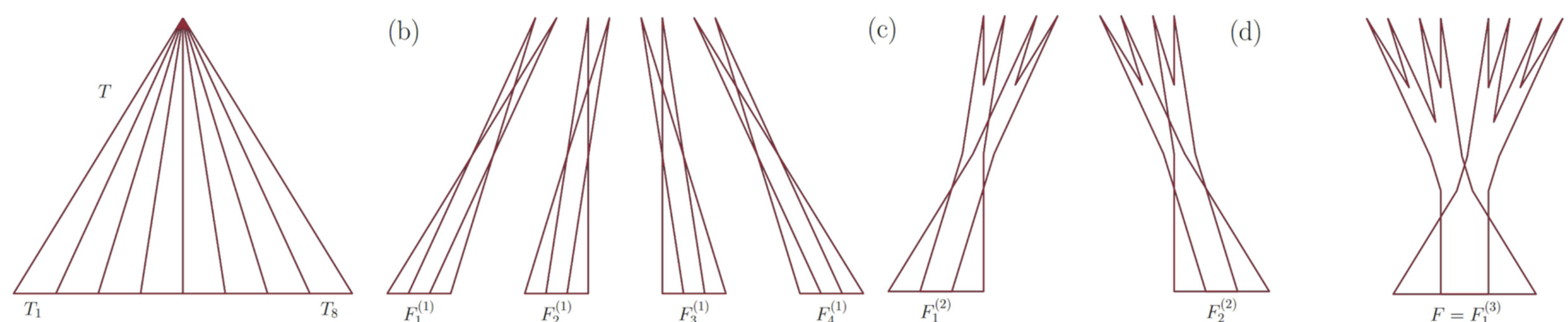


Figure 3. Diagram of a Perron tree

Theorem: Minimal Area ofakeya Sets in $\mathbb{R}^2$

There exists aakeya set K in $\mathbb{R}^2$ such that $\mathcal{L}^2(K) = 0$.

Starting with an equilateral triangle of height 1, at each iteration, F_i , we can find an open set U_i such that $F \subseteq U_i \subseteq U_{i-1}$ and as for each i , $\mathcal{L}^2(F_i) \leq 2^{-i}$, we have

$$\mathcal{L}^2(\overline{U_i}) \leq 2F_i \leq 2^{-i+1}.$$

We let $E = \bigcup_{i=1}^{\infty} \overline{U_i}$. This has measure zero, because for all i , we have $\mathcal{L}^2(E) \leq \mathcal{L}^2(\overline{U_i}) \leq 2^{-i+1}$. Let E' and E'' be two copies of E , rotated π and 2π degrees. Then the union of the three sets will contain a unit line segment of every direction, and have a measure of zero, thus making $K = E \cup E' \cup E''$ aakeya Set.

Dimension ofakeya Sets & Theakeya Conjecture

Theorem: Hausdorff Dimension ofakeya Sets in $\mathbb{R}^2$

Everyakeya set in $\mathbb{R}^2$ has a Hausdorff dimension equal to 2.

Let $K \subset \mathbb{R}^2$ be aakeya set. If $\mathcal{L}^2(K) > 0$, then it is trivial that $\dim(K) = 2$, by the definition of the Hausdorff dimension. Hence, we only need to consider when $\mathcal{L}^2(K) = 0$.

We use a ‘duality’ technique for this proof. Imagine all line segments contained within K , each with different slopes and x, y -intercepts. For each of these line segments, we associate them to a point in $\mathbb{R}^2$ as follows: for $q \in \mathbb{Q}$, the segment of the line $y = a(x - q) + b$ within K is represented by the point $(a, b) \in \mathbb{R}^2$. Then, using orthogonal projection maps and their properties, we can finish the proof.

Hausdorff dimensions essentially express how much the set ‘fills up’ the ambient space. The above result implies that althoughakeya sets in $\mathbb{R}^2$ can have zero measure, they still fill up the $\mathbb{R}^2$ space sufficiently that its Hausdorff dimension is equal to 2.

Theorem: Theakeya Conjecture

Everyakeya set in $\mathbb{R}^n$ has a Hausdorff dimension equal to n .

This is a more general version of the statement proved above. The $n = 3$ case was solved very recently in 2025, but $n \geq 4$ cases still remain as open problems. Theakeya conjecture is intimately connected to the *restriction conjecture*, another important unsolved problem in harmonic analysis.

akeya Needle Sets

Now we turn toakeya needle sets, where we let the unit line segment to translate and rotate continuously 360° inside the set.

Theorem:akeya Needle Sets can be Arbitrarily Small

For any $\varepsilon > 0$, there exists aakeya needle set $K \subset \mathbb{R}^2$ such that $\mathcal{L}^2(K) < \varepsilon$.

The above theorem follows from a simple geometric lemma. Let L_1, L_2 be parallel lines in $\mathbb{R}^2$ and γ be a unit line segment on L_1 . We can construct a set $A \subset \mathbb{R}^2$ that contains L_1 and L_2 with arbitrarily small measure such that γ can move continuously inside A from L_1 to L_2 .

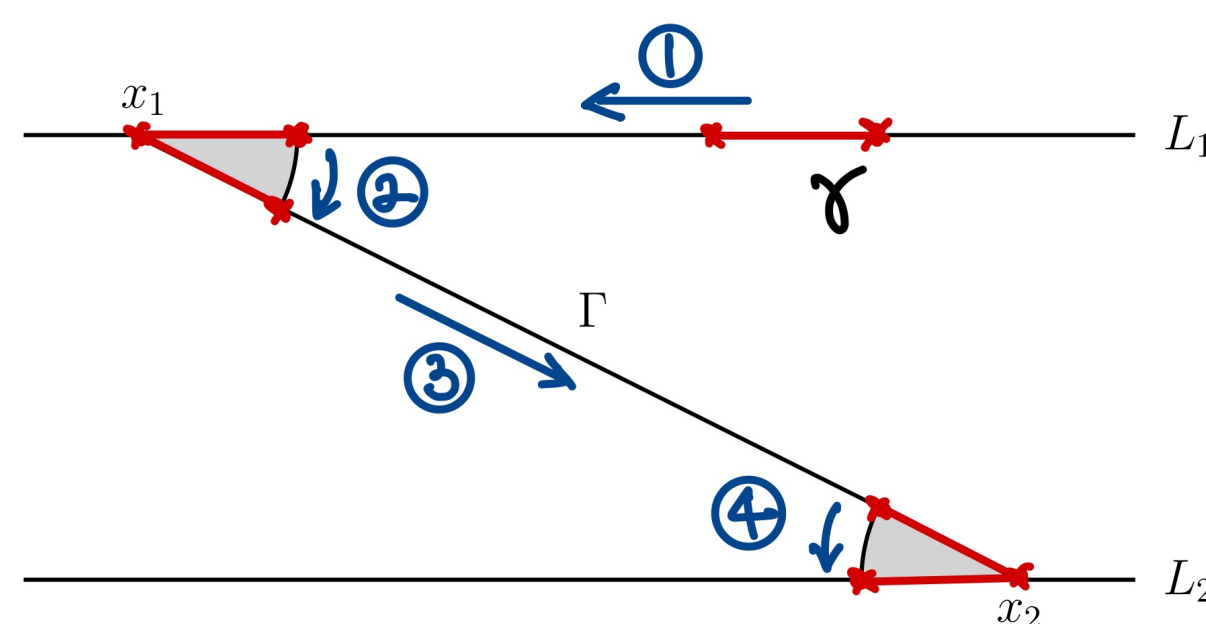


Figure 4. Continuous path of unit line segment γ from L_1 to L_2

Now recall the Perron tree construction that was used to generate aakeya set of measure zero. We go through the same process again, but this time, also add the gray ‘joints’ as in Figure 4, when we have a pair of parallel lines. Adding these joints allows a unit line segment in the set to freely translate and rotate inside the set, and we also have guarantee from the lemma that this does not affect the total measure. Hence, we can construct an arbitrarily smallakeya needle set.

References

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